Cross-species single-cell RNA-seq integration: from one-to-one orthologs to protein-language-model embeddings

A literature review of primary methods and recent benchmarks

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Methods surveyed	Span	Benchmarks	Widest divergence
15 primary tools	2018–2025	3 dedicated	>1.5 Gyr (TranscriptFormer)

Abstract

Comparative single-cell transcriptomics asks whether a cell type in one species has a homologue in another, and how its expression program has been conserved or rewired. The technical obstacle is that any two species' transcriptomes live in different gene coordinate systems. We review the three strategies that the field has used to bridge them: (i) subsetting to one-to-one orthologs and treating species as a batch, the recipe behind Seurat, LIGER, Harmony, Scanorama, BBKNN and the scVI family; (ii) replacing the hard ortholog filter with a soft, sequence-homology graph, as in SAMap; and (iii) discarding orthology entirely by embedding genes of every species into a common protein-language-model space, as in SATURN, UCE and TranscriptFormer. We summarise the primary methods papers, the evidence from three recent cross-species benchmarks, and the open problems that remain.

1 Problem statement

A cell atlas maps the cell types of an organism; a *comparative* atlas asks which of those types are shared across organisms and which are lineage innovations. Answering this at single-cell resolution requires placing cells from two or more species into one representation in which biological cell identity, and not the species of origin, drives the geometry. The obstacle is fundamental: transcriptomes are indexed by genes, and gene repertoires diverge. Orthologs are frequently many-to-many after lineage-specific duplication, lineage-specific genes have no counterpart at all, and even one-to-one orthologs can be co-opted into new expression programs. Integration methods differ chiefly in *how much* of this gene-level divergence they try to model, and that choice defines the three generations of tools reviewed here (Figure 1).

The first generation borrowed directly from within-species batch correction: after restricting both datasets to a shared one-to-one ortholog table, the species label is treated as just another batch covariate to be removed. This is convenient and, within mammals (~90 Mya divergence), largely defensible, because most genes still have a clean ortholog. It fails once the ortholog table itself becomes small or undefined — between vertebrates and invertebrates, or where no reference orthology exists. The second and third generations were built to reach those distances.

2 Ortholog-subsetting methods

Anchor- and factor-based aligners. Seurat v3 [1] finds correspondences by diagonalized canonical correlation analysis: it computes shared canonical-correlation vectors over the common-ortholog matrices, L2-normalises them, and calls mutual nearest neighbours in that subspace "anchors," which then drive a smooth correction and can transfer labels across datasets or species. LIGER [2] instead factorises each dataset by integrative non-negative matrix factorisation, $X_k \approx (W + V_k)H_k$, sharing metagene factors W across species while isolating dataset-specific signal in V_k. Harmony [3] operates post-PCA, iteratively soft-clustering cells and shifting centroids to maximise batch diversity within each cluster; its linear correction is fast and sensitive but, like every method in this section, can only recover signal that survives the ortholog bottleneck. Scanorama [4] and BBKNN [5] work at the graph level, stitching or rebalancing nearest-neighbour graphs across batches without an explicit generative model.

Conditional variational autoencoders. Deep generative models follow the same subset-then-batch recipe but absorb the species label into the decoder. scVI [6] fits a conditional VAE over the shared gene set, optimising the evidence lower bound

$$\mathcal{L}(\theta,\phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x},s)}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z}, s)\big] - D_{\mathrm{KL}}\big(q_\phi(\mathbf{z} \mid \mathbf{x}, s) \,\|\, p(\mathbf{z})\big), \tag{1}$$

where s is the species/batch indicator. Because s enters only the decoder, the latent $\mathbf{z}$ is, in principle, species-invariant. scANVI [7] extends Eq. (1) with a semi-

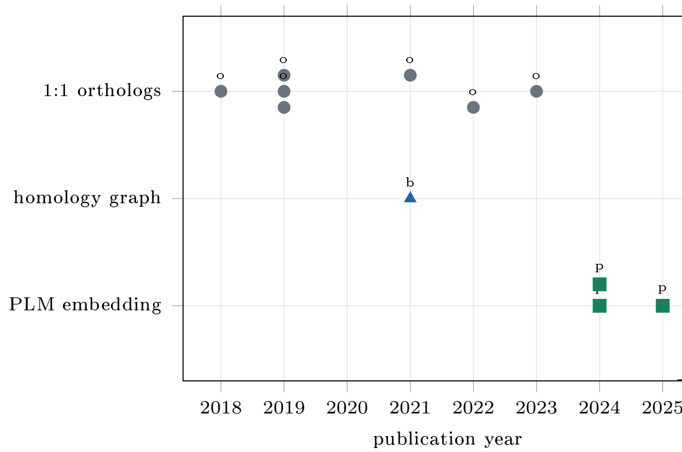

Figure 1: Three generations of cross-species integration. Primary methods positioned by publication year and by how they cross the gene-coordinate gap. Ortholog-subsetting tools (grey) dominate the early field because they double as batch-correction methods; the sequence-homology graph of SAMap (blue) and the protein-language-model embedders (green) are more recent and were built to span >400 Mya divergence, where a one-to-one ortholog table is small or undefined.

supervised label head, and scArches [8] adds lightweight adapter weights so a new species can be mapped onto a frozen reference without retraining. scPoli [9] augments the same backbone with learnable per-sample (or per-species) prototype embeddings, making the conditioning multi-scale rather than a single one-hot covariate. scGen [10] combines a VAE with explicit latent-space vector arithmetic and is a strong batch remover in the cross-species benchmarks below. The shared limitation is Eq. (1)'s support: every term is defined only on the intersection gene set, so lineage-specific and rewired genes are invisible by construction.

3 Homology-graph methods

SAMap [11] was the first tool to abandon the hard ortholog filter without abandoning gene identity. It replaces the fixed one-to-one table with a soft, BLAST-derived gene-homology weight matrix $\mathbf{H}$ and refines it iteratively against the cell-level manifold. Cross-species cell similarity is scored as

$$M_{ij} = \frac{\langle \mathbf{x}_i, \mathbf{H}\mathbf{y}_j \rangle}{\|\mathbf{x}_i\| \, \|\mathbf{H}\mathbf{y}_j\|}, \tag{2}$$

so that a gene in one species can contribute to a cell's profile in another through any of its sequence homologues — including paralogs, not only reciprocal best hits. Building on the self-assembling manifold algorithm, SAMap alternates between updating $\mathbf{H}$ and re-embedding cells until mutual cross-species neighbourhoods stabilise. This made integration across >500 Mya (sponge–planarian–schistosome –zebrafish) tractable for the first time, and, notably, the authors found many genes whose expression more closely tracks a paralog than the annotated ortholog, evidence that paralog substitution in cell-type programs is more common than a strict ortholog view assumes.

4 Protein-language-model embeddings

The most recent generation removes the orthology call entirely. Instead of matching genes, it embeds every gene — from any species — into a shared space defined by its protein sequence, using a protein language model such as ESM2.

SATURN. SATURN [12] takes scRNA-seq counts plus ESM2 protein embeddings for each species' genes, and learns "macrogenes" — soft clusters of functionally related proteins — that provide a common feature basis across datasets with disjoint gene sets. Cells are then mapped to a joint low-dimensional space using expression coupled to these protein representations. Because the shared basis is defined by sequence and not by an ortholog table, SATURN transfers annotations even between evolutionarily remote species and can flag genes that are co-expressed and functionally related across species without being orthologs.

Universal Cell Embeddings. UCE [13] pushes the idea to a self-supervised foundation model: a large transformer tokenises genes through their ESM2 protein embeddings and is trained, with no cell-type labels, on a corpus spanning human and seven other species. The resulting encoder maps any cell — including cells from species absent from training — into a single latent space with no fine-tuning, and was used to build an Integrated Mega-scale Atlas of tens of millions of cells across eight species. Its zero-shot, label-free operation is the practical payoff of the protein-space approach.

TranscriptFormer. TranscriptFormer [14] is a generative foundation model that jointly models genes and transcript counts with expression-aware attention, trained on up to 112 million cells spanning twelve species and roughly 1.5 billion years of evolution. In zero-shot settings it classifies cell types robustly even for species separated by hundreds of millions of years, and, being generative, it can be prompted to predict cell-type-specific transcription factors and gene–gene interactions, extending cross-species integration from alignment toward hypothesis generation.

5 Benchmarks and evaluation

Method proliferation has been matched by three benchmarking efforts of increasing cross-species scope. The scIB study of Luecken et al. [15] is not cross-species per

Table 1: Recommended method class by evolutionary distance, synthesising the cross-species benchmarks [16, 17].

Divergence	**Ortholog table**	**Best-supported class**
Intra-mammal ($\lesssim$100 Mya)	dense 1:1	conditional VAE / anchor
Cross-class / vertebrate	thinning	sequence-aware (SAMap)
Cross-phylum (>400 Mya)	sparse/absent	PLM embedding (SATURN/UCE)

se but set the evaluation template the field reuses: 68 method/preprocessing combinations over 1.2 M cells and 13 atlas-level tasks, scored by a weighted combination of batch-removal and biological-conservation metrics. Its central finding — that aggressive scaling trades biological conservation for batch removal — recurs in every cross-species study.

BENGAL [16] was the first benchmark dedicated to the cross-species question. It factorised the problem into two orthogonal choices — how genes are mapped by homology (four schemes) and which integration algorithm is applied (ten methods) — giving 28 strategies across 16 tasks on vertebrate cell types with clear one-to-one correspondences, and it contributed a conservation metric tailored to preserving cell-type distinguishability. Its message is that the gene-mapping step is as consequential as the integration algorithm, a variable the ortholog-subsetting literature had treated as fixed.

The most recent and most divergent benchmark, Zhong et al. [17], evaluated nine methods across 20 species, 4.7 million cells and eight phyla — effectively the whole animal taxonomic hierarchy. Two conclusions stand out. First, methods that exploit gene *sequence* information (SATURN) best preserved biological variance across large taxonomic distances, while generative batch-correction models (e.g. scGen) were strongest at species mixing — the same conservation-versus-removal tension, now resolved differently by method class. Second, no single tool dominated: SATURN was the most consistent from cross-genus to cross-phylum, SAMap was strongest for atlas-level integration beyond the cross-family level, and conditional-VAE methods remained competitive only within or below the cross-class range. In short, the right tool is a function of evolutionary distance.

6 Discussion and open problems

The trajectory of the field is a steady weakening of the orthology assumption: from a hard one-to-one filter, to a soft sequence-homology graph, to a protein-sequence embedding in which orthology is never called. Each step widens the reachable evolutionary distance but also loosens the biological anchor that makes an alignment interpretable, which is why evaluation has become the central difficulty rather than an afterthought.

Three problems remain open. First, *ground truth*: cross-species benchmarks rely on expert-assigned homologous cell types, which are themselves hypotheses, so a method that "mixes" species well may be rewarded for erasing genuine divergence. Second, *sequence is not expression*: protein-language-model embeddings encode what a protein *is*, not where or when it is transcribed, so two sequence-similar genes with divergent regulation can be conflated — the very paralog-substitution signal that SAMap exposed. Third, *interpretability at scale*: foundation-model embeddings deliver zero-shot convenience but make it harder to attribute an alignment to specific conserved genes, exactly the biological readout comparative studies want. Progress will likely come from hybrids that keep the protein-space reach of SATURN and UCE while restoring an explicit, testable gene-level account of what has been conserved and what has been rewired.

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